# Thanh Phung Truong

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## Research Profile

Thanh Phung Truong (Phung) is currently pursuing the Ph.D. degree in Computer Science and Engineering at Chung-Ang University, where he previously received the M.E. degree in Computer Science and Engineering in 2022. He received the B.E. degree in Electronics and Telecommunications from Ho Chi Minh City University of Technology in 2018. From 2019 to 2020, he worked as an FPGA Engineer at VinSmart Research and Manufacture Joint Stock Company, Vingroup, Vietnam. From 2022 to 2023, he was a Data Scientist at Viettel Networks, Viettel Group, Vietnam. He has been with the Ultra-Intelligent Computing/Communication Laboratory at Chung-Ang University since 2020. His research interests include wireless communications, network architectures, quantum networks, and applied machine learning.

## Education

**Ph.D. in Computer Science and Engineering**, Chung-Ang University, Seoul, South Korea — Mar. 2023–Aug. 2026(Exp.)
Advisor: Prof. Sungrae Cho

**M.E. in Computer Science and Engineering**, Chung-Ang University, Seoul, South Korea — Mar. 2020–Feb. 2022
Advisor: Prof. Sungrae Cho

**B.E. in Electronics and Telecommunications**, Ho Chi Minh City University of Technology, Vietnam National University Ho Chi Minh City, Vietnam — 2014–2018

## Research and Professional Experience

**Research Assistant**, Ultra-Intelligent Computing/Communication Laboratory, Chung-Ang University — 2020–Present

**Data Scientist**, Viettel Networks, Viettel Group, Vietnam — 2022–2023

**FPGA Engineer**, VinSmart Research and Manufacture Joint Stock Company, Vingroup, Vietnam — 2019–2020

## Research Interests

- Wireless communication
- Network architectures
- Quantum networks
- Applied machine learning

## Honors, Scholarships, and Awards

**Scholarships and Research Awards**

- **CAU Alumni Scholarship**, Chung-Ang University Graduate School Alumni Association, 2026.
- **Excellent in Research Award, CSE, CAU** (2023, 2024).
- **Chung-Ang University Young Scientist Scholarship**, 2023.
- **Chung-Ang University Young Scientist Scholarship**, 2020.

**Paper Awards**

- **Best Paper Awards:** ICOIN 2026, ICIT 2023, ICTC 2022.
- **Excellent Paper Awards:** ICMIC 2025, ICAIIC 2025.
- **Outstanding Paper Award:** ICMIC 2023.

## Publications

### Journal Articles

1. **T. P. Truong**, N.-P. Tran, J. Oh, D. Lee, V. D. Tuong, N.-N. Dao, and S. Cho, "Energy Efficiency in Quantized MIMO-NOMA Communication Systems," *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 6629–6642, March 2026, doi: 10.1109/TCCN.2024.3512847.
2. A.-T. Tran, **T. P. Truong**, D. Won, N.-N. Dao, and S. Cho, "Weighted Sum-Rate Maximization in Rate-Splitting MISO Downlink Systems," *IEEE Transactions on Network Science and Engineering*, vol. 13, pp. 5522–5538, January 2026, doi: 10.1109/TNSE.2024.3498735.
3. T. S. Do, **T. P. Truong**, Q. T. Do, D. Won, W. Noh, and S. Cho, "BidDeepSC-1.58b: 1.58-bit Bidirectional Slimmable Semantic Communication System," *IEEE Transactions on Vehicular Technology*, to appear, 2026, doi: 10.1109/TVT.2024.3511426.
4. T. H. Moges, **T. P. Truong**, D. S. Lakew, T. H. Huong, V. T. Hoang, N.-N. Dao, and S. Cho, "Age of Information-Aware Trajectory Optimization for Time-Sensitive UAV Systems in Uplink SCMA Networks," *Computer Networks*, vol. 268, no. 111369, August 2025.
5. T. S. Do, **T. P. Truong**, Q. T. Do, and S. Cho, "TranGDeepSC: Leveraging ViT Knowledge in CNN-based Semantic Communication System," *ICT Express*, vol. 11, no. 2, pp. 335–340, April 2025.
6. **T. P. Truong**, T. M. T. Nguyen, T. V. Nguyen, N.-N. Dao, and S. Cho, "Energy Efficiency in RSMA-Enhanced Active RIS-Aided Quantized Downlink Systems," *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 3, March 2025, doi: 10.1109/JSAC.2024.3521406.
7. **T. P. Truong**, H. V. Nguyen, N.-N. Dao, W. Noh, and S. Cho, "Orthogonalized RSMA based Flexible Multiple Access in Digital Twin Edge Networks," *IEEE Transactions on Wireless Communications*, vol. 23, no. 12, December 2024, doi: 10.1109/TWC.2024.3468791.
8. T.-H. Nguyen, **T. P. Truong**, A.-T. Tran, N.-N. Dao, L. Park, and S. Cho, "Intelligent Heterogeneous Aerial Edge Computing for Advanced 5G Access," *IEEE Transactions on Network Science and Engineering*, vol. 11, no. 4, July–August 2024, doi: 10.1109/TNSE.2024.3369045.
9. N.-P. Tran, **T. P. Truong**, Q. T. Do, N.-N. Dao, and S. Cho, "Adaptive Content Delivery and Bitrate Adaptation for QoE Improvement in IRS-Aided RSMA-Enabled IoMT Streaming Systems," *Internet of Things*, vol. 25, no. 101145, April 2024.
10. **T. P. Truong**, V. D. Tuong, N.-N. Dao, and S. Cho, "FlyReflect: Joint Flying IRS Trajectory and Phase Shift Design using Deep Reinforcement Learning," *IEEE Internet of Things Journal*, vol. 10, no. 5, pp. 4605–4620, March 2023, doi: 10.1109/JIOT.2022.3220581.
11. T.-V. Nguyen, **T. P. Truong**, T. M. T. Nguyen, W. Noh, and S. Cho, "Achievable Rate Analysis of Two-Hop Interference Channel with Coordinated IRS Relay," *IEEE Transactions on Wireless Communications*, vol. 21, no. 9, pp. 7055–7071, September 2022, doi: 10.1109/TWC.2022.3155746.
12. **T. P. Truong**, N.-N. Dao, and S. Cho, "HAMEC-RSMA: Enhanced Aerial Computing Systems with Rate Splitting Multiple Access," *IEEE Access*, vol. 10, pp. 52398–52409, May 2022, doi: 10.1109/ACCESS.2022.3174621.
13. V. D. Tuong, **T. P. Truong**, T.-V. Nguyen, W. Noh, and S. Cho, "Partial Computation Offloading in NOMA-Assisted Mobile Edge Computing Systems Using Deep Reinforcement Learning," *IEEE Internet of Things Journal*, vol. 8, no. 17, pp. 13196–13208, September 2021, doi: 10.1109/JIOT.2021.3062481.

### Selected Conference Papers

1. **T. P. Truong**, A.-T. Tran, V. D. Tuong, N.-N. Dao, and S. Cho, "NOMA-Enhanced Quantized Uplink Multi-user MIMO Communications," in *Proc. of IEEE INFOCOM*, 2024 (Acceptance Rate: 19.6%).
2. J. Oh, D. Won, **T. P. Truong**, and S. Cho, "Communication-Efficient Federated Learning with Local-Reconstruction Error-Feedback-Based Rescaled 1-Bit Compressive Sensing," in *Proc. of IEEE ICC*, Glasgow, Scotland, UK, May 2026.
3. **T. P. Truong**, T. M. T. Nguyen, T. V. Nguyen, N.-N. Dao, and S. Cho, "RSMA for Uplink MIMO Systems: DRL-based Achievable System Sum Rate Maximization," in *Proc. of IEEE GLOBECOM Workshops*, 2023.
4. T. S. Do, **T. P. Truong**, H. H. Tran, K. N. T. Tran, M. C. Ho, A. T. Tran, V. N. Vu, and S. Cho, "Multi-Task Semantic Communication System for DNA Transmission," in *Proc. of ICOIN*, Hanoi, Vietnam, January 2026 (Best Paper Award).

5. **T. P. Truong**, T. S. Do, J. Kim, S. Choi, J. Oh, and S. Cho, "Efficient Offloading for Edge Computing in UAV-Assisted Maritime Networks," in *Proc. of ICMIC*, Cebu, the Philippines, August 2025 (Excellent Paper Award).
6. T. S. Do, **T. P. Truong**, Q. T. Do, D. Won, A. B. Wondmagegn, and S. Cho, "Super-Resolution Semantic Communication System for Satellite Image," in *Proc. of ICAIIC*, Fukuoka, Japan, February 2025 (Excellent Paper Award).
7. **T. P. Truong**, T.-H. Nguyen, A.-T. Tran, S. V.-T. Tran, V. D. Tuong, L. V. Nguyen, W. Na, L. Park, and S. Cho, "Deep Reinforcement Learning-Based Sum-Rate Maximization for Uplink Multi-user SIMO-RSMA Systems," in *Proc. of ICIT*, Ho Chi Minh City, Vietnam, October 2023 (Best Paper Award).
8. **T. P. Truong**, N.-P. Tran, C. M. Ho, T. T. H. Pham, A.-T. Tran, and S. Cho, "Multiple Access in HAP-Enabled Maritime IoT Networks: A Deep Reinforcement Learning Approach," in *Proc. of ICMIC*, Jeju, South Korea, August 2023 (Outstanding Paper Award).
9. T.-H. Nguyen, **T. P. Truong**, N.-N. Dao, W. Na, H. Park, and L. Park, "Deep Reinforcement Learning-based Partial Task Offloading in High Altitude Platform-aided Vehicular Networks," in *Proc. of ICTC*, Jeju Island, South Korea, 2022 (Best Paper Award).

## References

- Prof. Sungrae Cho, Chung-Ang University (srcho@cau.ac.kr).
- Prof. Nhu-Ngoc Dao, Sejong University (nndao@sejong.ac.kr).